

Day 11: Inference for MLR

MLR model for `bikes` data

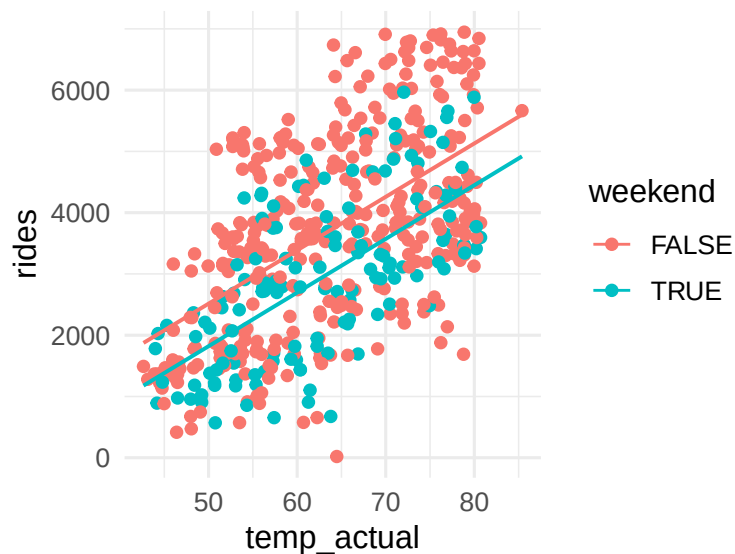
We're back to the separate-lines model from last class:

term	estimate	std. error	t	p-value
Intercept	-1843.867	415.700	-4.44	<0.001
temp_actual	87.173	6.439	13.54	<0.001
weekendTRUE	-714.938	795.698	-0.90	0.369
temp_actual:weekendTRUE	0.411	12.539	0.03	0.974

💡 Warm up

Write out the theoretical and fitted regression model for `rides ~ temp_actual * weekend`.

Do we need a different-slopes model?



Look at the `temp_actual : weekendTRUE` row in the table above – its estimate is tiny and its p-value is large, suggesting the slopes might not really differ.

Testing a single coefficient

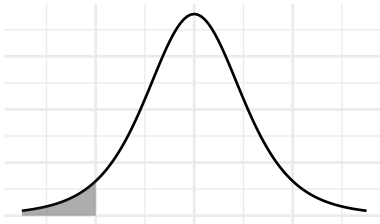
Hypotheses: $H_0 : \beta_j = \#$ vs. $H_a : \beta_j \neq, <, \text{ or } > \#$

Test statistic: $t = \frac{\hat{\beta}_j - \#}{SE(\hat{\beta}_j)}$

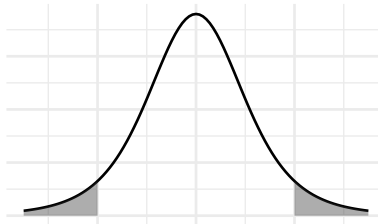
Reference distribution: t distribution with $df = n - (k + 1)$, where k = number of predictors in the model

p-value: area in the tail(s) specified by H_a

$H_a : \beta_j < 0$



$H_a : \beta_j \neq 0$



$H_a : \beta_j > 0$



CI for a single coefficient: $\hat{\beta}_j \pm t_{n-(k+1)}^* \cdot SE(\hat{\beta}_j)$

Do we need weekend at all?

Careful – with the interaction term still in the model, the weekendTRUE row alone doesn't answer “do we need weekend at all,” because weekend also appears in the interaction term.

term	estimate	std. error	t	p-value
Intercept	-1843.867	415.700	-4.44	<0.001
temp_actual	87.173	6.439	13.54	<0.001
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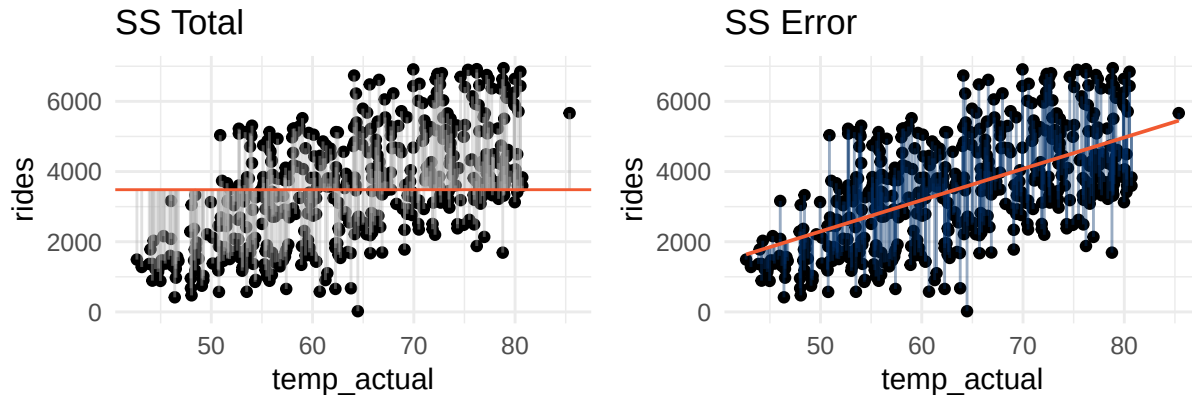
Caution!

A single coefficient's p-value only tests that ___ term, holding all others in the model. To ask whether weekend matters *at all*, we need to test $\beta_2 = \beta_3 = 0$ jointly – more on this below.

Once we drop the (non-significant) interaction, the simpler parallel-lines model shows weekend is clearly still useful:

term	estimate	std. error	t	p-value
Intercept	-1850.776	357.953	-5.17	<0.001
temp_actual	87.281	5.520	15.81	<0.001
weekendTRUE	-689.179	124.639	-5.53	<0.001

Explained variability



Idea: does a larger model explain a *significantly* bigger share of the total variability than a smaller model?

Can we just compare R^2 values?

Model	R^2	Adjusted R^2
temp only	0.3322	0.3308
temp + weekend	0.3709	0.3683
temp + windspeed + weekend	0.3898	0.3861
temp + windspeed*weekend	0.3901	0.3852

$$R^2 = 1 - \frac{SSE}{SST} \quad R^2_{adj} = 1 - \frac{SSE/(n-k)}{SST/(n-1)} = 1 - \frac{MSE}{MST}$$

- R^2 **always** increases when predictors are added, even useless ones
- R^2_{adj} **could** _____ if unnecessary predictors are added
- Better to use R^2_{adj} for comparing multiple regression models

Formal test for comparing models

	Model
Full	$\mu\{Y X\} = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1x_2$
Reduced	$\mu\{Y X\} = \beta_0 + \beta_1x_1$

Hypotheses: $H_0 : \beta_2 = \beta_3 = 0$ vs. $H_a : \text{at least one } \beta_j \neq 0, j = 2, 3$

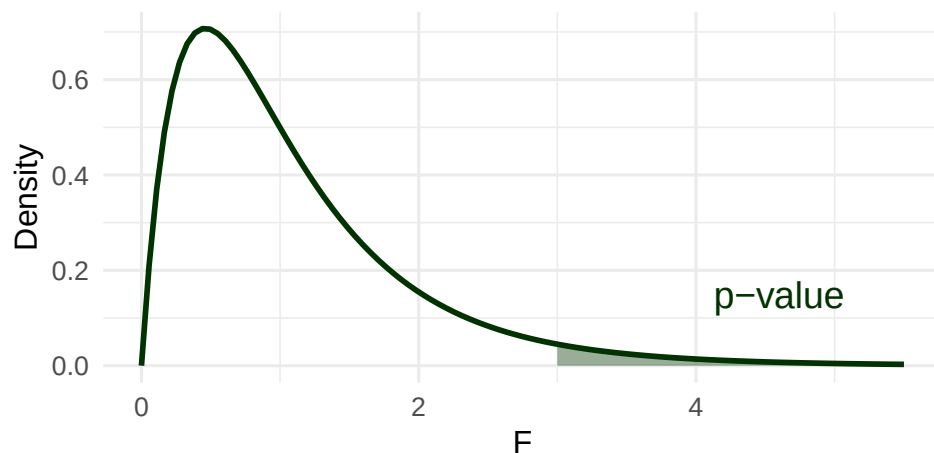
Comparing models with an F-test

$$F = \frac{(R_{\text{full}}^2 - R_{\text{reduced}}^2)/d}{(1 - R_{\text{full}}^2)/df_{\text{full}}} = \frac{(SSR_{\text{full}} - SSR_{\text{reduced}})/d}{MSE_{\text{full}}} = \frac{(SSE_{\text{reduced}} - SSE_{\text{full}})/d}{MSE_{\text{full}}}$$

where

- $d = df_{\text{full}} - df_{\text{reduced}} = \# \text{ of } \beta\text{'s being tested}$
- $df_i = n - (p + 1) = \text{error d.f. for model } i$
- $MSE_{\text{full}} = \frac{SSE_{\text{full}}}{df_{\text{full}}}$

The F statistic follows an F distribution with d and $n - p - 1$ degrees of freedom. Like the χ^2 , it's right-skewed and only takes positive values, so we only ever look at the **upper tail** for a p-value:



Upper-tail area in R: `1 - pf(stat, df1, df2)`

Putting it all together

model	r.squared	df	df.residual	nobs
Full: temp*weekend	0.371	3	496	500
Reduced: temp only	0.332	1	498	500

$$F = \frac{(R_{\text{full}}^2 - R_{\text{reduced}}^2)/d}{(1 - R_{\text{full}}^2)/df_{\text{full}}} = \frac{(0.371 - 0.332)/2}{(1 - 0.371)/496} \approx 15.3$$

p-value = 3.71e-07 – extremely small, so we _____ H_0 : adding weekend (and its interaction with temp_actual) meaningfully improves the model.

Doing it directly in R with `anova()`

`anova(reduced, full)` takes the **reduced** model first, then the **full** model, and runs exactly the F-test above – no need to compute R^2 's by hand.

Analysis of Variance Table

Model 1: rides ~ temp_actual

Model 2: rides ~ temp_actual * weekend

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	498	826326178				
2	496	778436469	2	47889709	15.257	3.713e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1